

Understanding the Walert RAG Pipeline

Walert RAG Pipeline understanding for Milestone 1, Victorian Road Rules RAG Assistant

Source: github.com/rmit-ir/walert (README.md, COMBINED_README.md)

What Walert Does:

Walert is a conversational agent built by RMIT to answer FAQ's about programs of study at the School of Computing Technologies. It uses Retrieval Augmented Generation - RAG, rather than answering purely from an LLM's memory, it fetches the most relevant passages from an organized knowledge base first, then generates an answer grounded in that content.

The Pipeline:

Walert takes FAQ content and turns it into a knowledge base of individual passages (Collection.csv), each with a unique ID, representing the exact text extracted from an FAQ answer. Separately, it stores the actual questions people might ask (topics.csv), where each of the 43 underlying topics has 1 to 4 different rephrasings, generated using the Falcon LLM to reflect how real users might word things differently.

When a question comes in, the system searches this passage knowledge base using one of two methods:

- BM25 — a traditional keyword-matching search
- DPR (Dense Passage Retrieval) — a neural embedding-based semantic search

Both are paired with Falcon for generation, allowing a direct comparison between a classic and a neural retrieval approach.

If only one passage matches the question, it's used directly. If a question needs information from multiple passages, those passages are combined and summarized together (stored in gold_summaries.csv) into one clear answer. Finally, Falcon(LLM) generates the response to the user, grounded in whichever passages were retrieved.

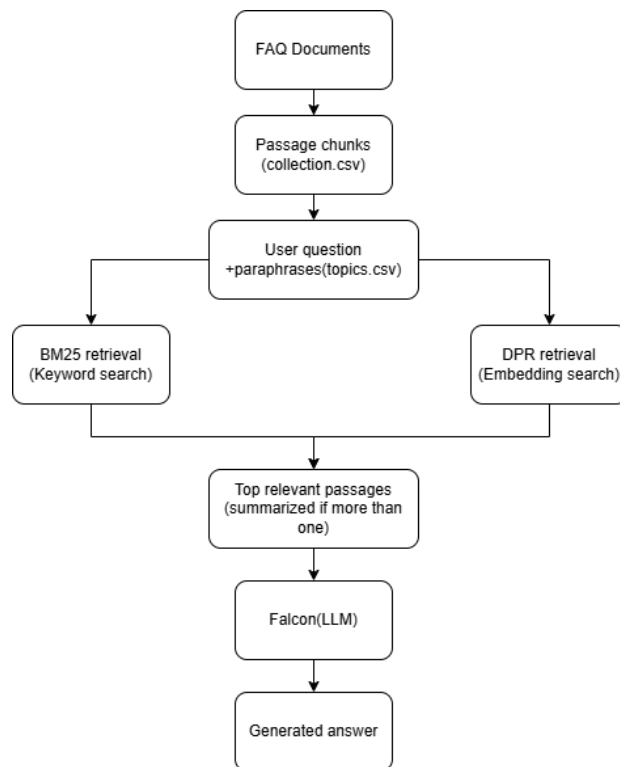
Ground truth for evaluation is stored in groundtruth.csv and qrels.txt, mapping each topic or question to its correct passages with a relevance judgment score of 1 (partial) or 2 (full relevance).

Pipeline Flow:

FAQ documents

- > Passage chunks (Collection.csv)
- > User question + paraphrases (topics.csv)
- > Retrieval: BM25 (keyword) OR DPR (embedding)
- > Top relevant passages - summarized if more than one (gold_summaries.csv)
- > Falcon (LLM)
- > Generated answer

Flowchart:



What We Can Reuse for the Victorian Road Rules Assistant

Walert stage	How it maps to our project
FAQ documents-> Passages (Collection.csv)	Chunk official VicRoads rule documents into individual passages
Topics + paraphrased questions(topic.csv)	Write 2-4 phrasings per road rule question
BM25 vs DPR retrieval comparison	Reproduce this same comparison on our road-rules knowledge base as an evaluation
Groundtruth.csv / qrels.txt relevance mapping	Reusable pattern for our own gold-standard test set (Member 4's evaluation work)
gold_summaries.csv multi passage summarization	Useful when a question spans multiple rules (e.g.: demerit points differing by license type)

References

Cherumanal, S. P., Tian, L., Abushaqra, F. M., Magnossão de Paula, A. F., Ji, K., Ali, H., Hettiachchi, D., Trippas, J. R., Scholer, F., & Spina, D. (2024). Walert: Putting Conversational Information Seeking Knowledge into Action by Building and Evaluating a Large Language Model-Powered Chatbot. CHIIR '24.

GitHub repository: <https://github.com/rmit-ir/walert>